

Flagged LLT Polynomials

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ICERM ECR Seminar

joint work with Jonah Blasiak, Mark Haiman, Jennifer Morse, and Anna Pun

18 November 2025

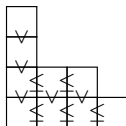
Young Tableaux

Definition

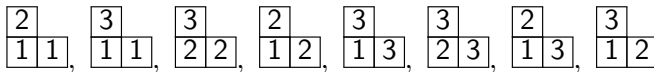
Filling of partition diagram of λ with numbers such that

- 1 strictly increasing up columns
- 2 weakly increasing along rows

Collection is called $\text{SSYT}(\lambda)$.



For $\lambda = (2, 1)$,



Schur polynomials

Associate a polynomial to $\text{SSYT}(\lambda)$.

$$\begin{array}{cccccccc} \begin{array}{|c|c|} \hline 2 & \\ \hline 1 & 1 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 3 & \\ \hline 1 & 1 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 3 & \\ \hline 2 & 2 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 2 & \\ \hline 1 & 2 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 3 & \\ \hline 1 & 3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 3 & \\ \hline 2 & 3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 2 & \\ \hline 1 & 3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 3 & \\ \hline 1 & 2 \\ \hline \end{array} \\ \rightarrow & \begin{array}{|c|c|} \hline z_2 & \\ \hline z_1 & z_1 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_3 & \\ \hline z_1 & z_1 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_3 & \\ \hline z_2 & z_2 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_2 & \\ \hline z_1 & z_2 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_3 & \\ \hline z_1 & z_3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_3 & \\ \hline z_2 & z_3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_2 & \\ \hline z_1 & z_3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_3 & \\ \hline z_1 & z_2 \\ \hline \end{array} \end{array}$$

$$s_{(2,1)}(z_1, z_2, z_3) = z_1^2 z_2 + z_1^2 z_3 + z_2^2 z_3 + z_1 z_2^2 + z_1 z_3^2 + z_2 z_3^2 + 2z_1 z_2 z_3$$

Definition

For λ a partition

$$s_\lambda = \sum_{T \in \text{SSYT}(\lambda)} \mathbf{z}^T \text{ for } \mathbf{z}^T = \prod_{i \in T} z_i$$

- s_λ is a symmetric function.
- $\{s_\lambda\}_\lambda$ forms a basis for Λ .

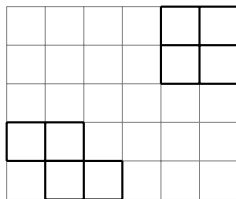
Products of Schur polynomials

- Littlewood-Richardson rule: $s_\lambda s_\mu = \sum_\nu c_{\lambda\mu}^\nu s_\nu$ for $c_{\lambda\mu}^\nu \in \mathbb{N}$.
- Skew Schur function: $s_{\nu/\lambda} = \sum_{T \in \text{SSYT}(\nu/\lambda)} x^T = \sum_\mu c_{\lambda\mu}^\nu s_\mu$.
- Straightforward observation: $s_{\nu/\lambda} s_{\kappa/\mu} = \sum_{\substack{T \in \text{SSYT}(\nu/\lambda) \\ S \in \text{SSYT}(\kappa/\mu)}} x^T x^S$.
- q -deformation?

Key Object: LLT Polynomials

Let $\nu = (\nu_{(1)}, \dots, \nu_{(k)})$ be a tuple of skew shapes. (Skew shape = $\lambda \setminus \mu$)

$$\nu = \left(\begin{array}{|c|c|c|} \hline \square & \square & \\ \hline \square & \square & \square \\ \hline \end{array}, \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \right)$$



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-4	-3	-2	-1	0	1
-3	-2	-1	0	1	2
-2	-1	0	1	2	3
-1	0	1	2	3	4
0	1	2	3	4	5

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- A *semistandard tableau* on ν is a map $T: \nu \rightarrow \mathbb{Z}_+$ which restricts to a semistandard tableau on each $\nu_{(i)}$.

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- $\mathcal{G}_\nu$ is Schur-positive for any tuple of skew shapes ν [Grojnowski-Haiman, 2007].

Some notable occurrences of LLT Polynomials

- Haglund-Haiman-Loehr formula for Macdonald polynomials:

$$\tilde{H}_{\mu}(X; q, t) = t^{n(\mu)} \sum_R \left(\prod_{\substack{\square \\ \overline{u}}} q^{a+1} t^l \right) \mathcal{G}_R(X; t^{-1}).$$

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- For G = incomparability graph for natural unit interval order (encoded by Dyck path P),

$$\chi_G(x; t) = (1 - t)^{-|V|} \mathcal{G}_{\nu(P)}[(1 - t)x; t].$$

Flagged Tableaux

- Fix partition shape $\lambda = (\lambda_1, \dots, \lambda_l)$ and *flag*
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- Denote the set of such tableaux via $\text{FT}(\lambda, \mathbf{b})$.
- E.g., $\lambda = (2, 1)$, $\mathbf{b} = (1, 3)$, $\text{FT}(\lambda, \mathbf{b}) =$

$$\begin{array}{|c|c|} \hline 2 & \leq 3 \\ \hline 1 & 1 \\ \hline \end{array} \leq 1 \quad \begin{array}{|c|c|} \hline 3 & \leq 3 \\ \hline 1 & 1 \\ \hline \end{array} \leq 1$$

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- Note, there is no issue with letting length of $\mathbf{b}$ be less than l .

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- $\mathfrak{S}_w(x) = s_{\lambda, \mathbf{b}}(x)$ for *vexillary* $w \in S_n$ (2143-avoiding).
- $s_{\lambda, \mathbf{b}}(x)$ are examples of *Demazure characters*.

Demazure characters and atoms

The *Demazure operator* π_i acts on $f \in \mathbb{Q}(q, t)[x_1^{\pm 1}, \dots, x_N^{\pm 1}]$ by

$$\pi_i(f) = \frac{x_i f - x_{i+1} s_i(f)}{x_i - x_{i+1}}.$$

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The *Demazure characters* or *key polynomials* are constructed from

- $\mathcal{D}_\lambda = x^\lambda := x_1^{\lambda_1} \cdots x_N^{\lambda_N}$ for partition λ .
- $\mathcal{D}_{s_i(\alpha)} = \pi_i \mathcal{D}_\alpha$ for $\alpha_i > \alpha_{i+1}$, for any $\alpha \in \mathbb{N}^N$.

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Demazure atoms are defined the same as keys but with $\hat{\pi}_i := \pi_i - 1$ in place of π_i :

- $\mathcal{A}_\lambda = x^\lambda$ for partition λ .
- $\mathcal{A}_{s_i(\alpha)} = \hat{\pi}_i \mathcal{A}_\alpha$ for $\alpha_i > \alpha_{i+1}$, for any $\alpha \in \mathbb{N}^N$.

Examples

$$\mathcal{D}_{520} = x_1^5 x_2^2$$

$$\mathcal{D}_{250} = \pi_1 \mathcal{D}_{520} = \pi_1(x_1^5 x_2^2) = x_1^5 x_2^2 + x_1^4 x_2^3 + x_1^3 x_2^4 + x_1^2 x_2^5$$

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$$\mathcal{D}_{520} = \mathcal{A}_{520}$$

$$\mathcal{D}_{250} = \mathcal{A}_{520} + \mathcal{A}_{250}$$

$$\mathcal{D}_{205} = \mathcal{A}_{520} + \mathcal{A}_{250} + \mathcal{A}_{502} + \mathcal{A}_{205}$$

$$\vdots$$

Recovering Symmetric Functions

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- For *Weyl symmetrization operator* π_{w_0} , we have

$$\pi_{w_0}(s_{\lambda, \mathbf{b}}(x)) = s_{\lambda}(x), \quad \pi_{w_0}(\mathcal{D}_{\alpha}) = \mathcal{D}_{\text{sort}(\alpha)} = s_{\alpha+},$$

$$\pi_{w_0}(\mathcal{A}_{\alpha}) = \begin{cases} s_{\alpha} & \alpha \text{ a partition,} \\ 0 & \text{else.} \end{cases}$$

Flagged LLT Polynomials (Blasiak-Haiman-Morse-Pun-S.)

- Let $e_1, \dots, e_l$ be the row ends of ν , ordered in reverse reading order.
- Fix flag $\mathbf{b} = (b_1 \leq \dots \leq b_l)$.
- $\text{FT}(\nu, \mathbf{b})$ = set of semistandard tableaux T on ν satisfying $T(e_i) \leq b_i$.
- The *flagged LLT polynomial* indexed by ν and $\mathbf{b}$ is

$$\mathcal{G}_{\nu, \mathbf{b}}(x; t) = \sum_{T \in \text{FT}(\nu, \mathbf{b})} t^{\text{inv}(T)} x^T.$$

						e_3
						e_1
	e_4					
		e_2				

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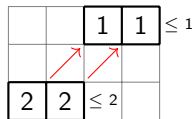
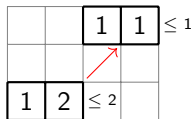
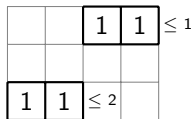
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2	4					≤ 4
	1	2				≤ 2

Flagged LLT Polynomials

$$\nu = (\square\square, \square\square), \mathbf{b} = (1, 2)$$

T



$$\mathcal{G}_{r,\nu}(x; t) = x_1^4 + t x_1^3 x_2 + t^2 x_1^2 x_2^2$$

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- $\mathcal{G}_{\nu, \mathbf{b}}(x; t)$ have an algebraic formula via “nonsymmetric Hall-Littlewood polynomials.”
- $\mathcal{G}_{\nu, \mathbf{b}}(x; t)$ are conjecturally positive in terms of *Demazure atoms*.

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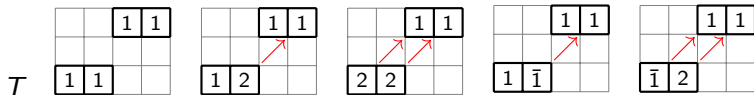
The *signed flagged LLT polynomial* indexed by ν and flag $\mathbf{b}$ is

$$\mathcal{G}_{\nu, \mathbf{b}}^\pm(\mathbf{x}; t) = \sum_{T \in \text{FT}^\pm(\nu, \mathbf{b})} t^{\text{inv}(T)} (-t)^{-\#\text{bar}(T)} \mathbf{x}^{|T|},$$

where $|T|$ is the result of removing all bars from T .

Signed flagged LLT polynomials

$$\nu = (\square\square, \square\square), \mathbf{b} = (1, 2)$$



$$\begin{aligned} \mathcal{G}_{\nu, \mathbf{b}}^{\pm}(\mathbf{x}; t) &= x_1^4 + tx_1^3x_2 + t^2x_1^2x_2^2 - x_1^4 - tx_1^3x_2 \\ &= t^2x_1^2x_2^2 \end{aligned}$$

Flagged plethysm

Define *flagged plethysm* $\Pi_{t,x}: \mathbb{K}[x_1, \dots, x_l] \rightarrow \mathbb{K}[x_1, \dots, x_l]$ to be the (over-determined) linear map

$$\Pi_{t,x} \mathcal{G}_{\nu, \mathbf{b}}^{\pm}(x; t^{-1}) = \mathcal{G}_{\nu, \mathbf{b}}(x; t^{-1}).$$

Theorem (Blasiak-Haiman-Morse-Pun-S., 2025+)

$\Pi_{t,x}$ is well-defined.

Note, $\Pi_{t,x}$ is a “nonsymmetric analogue” of the plethystic map $f[X] \mapsto f[X/(1-t)]$ for symmetric function f .

Applications

- ① Can define *modified r -nonsymmetric Macdonald polynomials* (or *modern Macdonald polynomials*) via a *flagged* Haglund-Haiman-Loehr formula:

$$H_{\eta|\lambda}(x; q, t) = t^n \sum_R \left(\prod_{\substack{U \\ \square}} q^{a+1} t^l \right) \mathcal{G}_{R, (b_1, \dots, b_r)}(x; t^{-1}).$$

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- 2 Using $H_{\eta|\lambda}$ to define a nonsymmetric analogue of ∇ , we show a “nonsymmetric shuffle theorem” expressed in terms of flagged LLT polynomials associated to flagged Dyck paths.
- 3 Tewari-Wilson-Zhang define chromatic nonsymmetric polynomials associated to $d \times d$ Dyck paths starting with r north steps, $\chi_{\mathbf{b}, \pi}$. We show

$$\chi_{(1, \dots, r), \pi}(x; t) = (1 - t)^{r-d} \mathcal{G}_{\nu(\pi), (1, \dots, r)}^{\pm}(x; t).$$